Erick **White**

Profile

Hardworking student seeking an opportunity to learn and grow in a team environment with exceptional interpersonal skills and an ability to communicate effectively.
Hoping to explore the aerospace field (with a focus on mathematical modeling and computation in astrodynamics) through both academia and industry.

Contact Details

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Education

University of Colorado Boulder

- Aerospace Engineering BS '26
  - GPA: 4.0
- Applied Mathematics BS '26
  - GPA: 4.0

  Computer Science Minor

Key Skills

Programming

- MATLAB
- Python
- Java

LaTeX

Languages

- English (native)
- Italian (native)
- Spanish (semi-fluent)

References

References available upon request.

Experience

Undergraduate Research Assistant – *Celestial and Spaceflight Mechanics Laboratory* **June 2025 – Present**

◇ Researched and developed software to integrate N-body dynamics with granular mechanics
◇ Simulated and analyzed systems of asteroids to better understand relevant physical properties

Undergraduate Research Assistant – *Lightning, Atmosphere, Ionosphere, and Radiation Belt Research Group* **August 2024 – May 2025**

◇ Researched and developed methods for mass-generating spectrograms from Van Allen Probe magnetometer data capable of clearly showing plasma wave events
◇ Investigated effectiveness of using machine learning models to detect plasma wave events in Earth's magnetosphere
◇ Conducted preliminary analysis into plasma wave MLT and L-shell distributions for comparison to theoretical models

Analysis Intern – *Omitron/NASA CARA* **May 2024 – August 2024**

◇ Continued work from summer 2023 internship researching and developing a new method of visualizing satellite conjunction events
◇ Wrote research paper on conjunction visualization for submission to 2025 AAS/AIAA Astrodynamics Specialist Conference (accepted)

Software Intern – *Omitron/NASA CARA* **June 2023 – August 2023**

◇ Developed new and expanded upon existing unit tests for the NASA CARA SDK
◇ Developed a new, highly-customizable visualization program for satellite conjunction events to be used by CARA in training and mission analysis

Communication and Publications

2025 AAS/AIAA Astrodynamics Specialist Conference – *Showcase and Comparison of Three Methods for Visualizing Near-Earth Satellite Conjunction Events* **August 2025**

◇ Published paper on satellite conjunction visualization at national-level conference as first author
◇ Created poster for conference presentation to a wide variety of academic and industry backgrounds (presented by internship mentor)

Invited Talk – *Applications of Computer Vision to Space and Plasma Physics* **April 2025**

◇ Selected to produce poster for CU Boulder undergraduate research symposium
◇ Presented results of magnetospheric plasma wave research to faculty, students, and public

2023 SIAM Front Range Student Conference - *We Put the "UN" in FUN: The Mathematical Guide to Saving the World* **March 2023**

◇ Presented results of mathematical modeling paper to peers and professors at regional conference

Leadership

Outreach Chair – *University of Colorado Boulder Astronomy Club* **Fall 2023 – Present**

◇ Coordinated events with local schools and organizations to boost community involvement and interest in astronomy
◇ Facilitated organization and execution of club events, including stargazing outings, panel nights, and study sessions